

Fatimah Zahra

Computer Science Student | Software Engineering, Web Development, Data Science

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SUMMARY

Final year Computer Science student at Monash University Malaysia with experience developing web and mobile applications, interactive software and machine learning systems. My projects have involved working across the stack, from user interfaces and APIs to databases and data processing. I enjoy learning new technologies through hands-on development and applying them to build practical software.

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, JavaScript, SQL, C, TypeScript

Frameworks & Libraries: Flask, Vue.js, Node.js, React Native, PyTorch, Numpy, Pandas

Tools & Platforms: Git, VS Code, IntelliJ, Jupyter Notebook, Oracle, MongoDB, Figma

EXPERIENCE

Research Intern - School of IT Research Programme (iSR)

Nov 2025 - Feb 2026

Monash University Malaysia

- Built an end-to-end multimodal emotion recognition pipeline on the IEMOCAP dataset, with audio and text preprocessing in Python, Librosa, PyTorch and scikit-learn
- Evaluated SVM, BiLSTM, BERT and RoBERTa models under speaker-independent testing, achieving a **74.07%** weighted F1-score with a RoBERTa-BiLSTM hybrid at under 1 second inference per sample

Frontend Developer & UI/UX Designer - Software Engineering Process & Management Unit

July 2025 - Nov 2025

Monash University Malaysia | [GitHub](#)

- Built dynamic lesson pages, progress tracking and secure role-based interfaces in Vue.js, integrating with Node.js/Express backend services in a cross-functional team developing a full-stack LMS on the MEVN stack
- Applied Scrum practices, including sprint planning and code reviews, to improve team productivity and code quality

Web Developer - MIND ENGINE EXPO 2025

May 2025 - Aug 2025

Monash University Malaysia

- Built responsive frontend components and improved UI consistency across event pages for a large-scale Engineering & IT career fair website, in collaboration with the development team
- Coordinated with publicity and events teams to keep content accurate and up to date

PROJECTS

Anaesthesia Medical Toolkit | React Native, Expo, TypeScript

[GitHub](#)

- Designed and developed a cross-platform mobile application providing anaesthetic clinical references, drug dosage calculators, and Difficult Airway Society (DAS) decision support for healthcare professionals
- Built an intuitive React Native interface with interactive checklists, dynamic clinical algorithms, searchable drug references, and reusable component architecture using TypeScript and Expo
- Implemented state management, custom navigation, responsive layouts and polished UI interactions to improve usability in fast-paced clinical environments

Garbage Inc. | Java, OOP, UML

[GitHub](#)

- Developed a multiplayer survival game inspired by the well-known indie game Lethal Company, focusing on cooperative gameplay
- Implemented core systems such as enemy behaviours, item interactions, and environment hazards using OOP design principles
- Designed modular game architecture using abstraction and polymorphism to support scalable gameplay features

Flappy Birb | JavaScript, TypeScript, RxJS

[GitHub](#)

- Implemented a browser-based Flappy Bird clone using Functional Reactive Programming (FRP) principles and observables for real-time game state management
- Developed core game mechanics including pipe collision detection, scoring, and ghost bird logic that replays previous runs
- Enhanced gameplay with a custom additional feature that changes the bird's colour after consecutive wins, improving user engagement and visual feedback

GridPix | Flask, Python, SQL, HTML/CSS, JavaScript

[GitHub](#)

- Developed a full-stack pixel art web application with interactive drawing tools and persistent artwork storage
- Implemented user authentication and customisable grid functionality using Flask and SQL databases

EDUCATION

Bachelor of Computer Science - Monash University Malaysia

Jul 2024 - Present

Relevant Coursework: Web Development, Object-Oriented Design, Data Structures and Algorithms, Programming Paradigms, Databases, Artificial Intelligence, Data Science

LEADERSHIP

Head Of Secretary - Google Developer Group on Campus (GDGoC)

Dec 2025 - Nov 2026

- Managed meeting coordination, documentation, and internal communications to support smooth execution of club activities and technical events
- Developed registration and feedback workflows using Google Forms to streamline participant management and event coordination

Publicity Officer - Monash University Malaysia Tech Club (MUMTEC)

Jan 2025 - Nov 2025

- Led branding initiatives through website design, logo creation, and promotional content development
- Produced marketing materials and maintained consistent visual identity across social media and event platforms